

Detailed Software Requirements Specification (DSRS) for AI-Powered Student Feedback Analyzer for Curriculum Enhancement

Think back

Group 31

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Document Revisions

Date	Version	Description	Author

Document Approval

I have reviewed this document and confirm that the contents are accurate and complete. Any future changes to this document must be communicated in writing and approved by me as the supervising lecturer/coordinator.

Signature
Date:
Name:

1. INTRODUCTION

1.1 Purpose

This document specifies the Software Requirements Specification (SRS) for the AI-Powered Student Analyzer and Curriculum Suggestion System for Curriculum Enhancement. Its main purpose is to clearly describe what the system should do, how it should work, and what features it will include before we begin designing and building it.

It will help the development team understand what needs to be created and guide them during design, development, and testing. It will also help stakeholders understand the system's goals, scope, and capabilities.

Once this document is approved, any changes to the system requirements will be recorded, including how they might affect the project's cost, scope, or schedule, and they will be shared with the client for approval.

1.2 Summary

In the current system, the process of collecting, managing, and analyzing feedback is frequently inefficient, fragmented, and lacking in automation. Users and stakeholders face difficulties in submitting feedback, tracking responses, and ensuring that their input leads to meaningful improvements. In addition, administrators and decision makers struggle to manually process large volumes of feedback and derive actionable insights.

This project aims to address these challenges by creating an automated feedback management system that streamlines the entire feedback lifecycle, from submission and categorization to analysis and reporting. By implementing a centralized, user-friendly platform, the system will improve communication between users and administrators, improve data accuracy, and support data-driven decision-making. Ultimately, this solution will increase user satisfaction, improve response times, and contribute to continuous improvement within the organization.

1.3 Company Overview

The AI-Powered Student Feedback Analyzer and Curriculum Suggestion System is being developed by a team of final-year students as part of the EEY6689 Final Year Group Project (Software Engineering) course. The system serves as an intelligent digital platform designed to help educational institutions collect, analyze, and understand student feedback more effectively using Artificial Intelligence (AI).

The development team functions as a software solution provider focused on improving the quality of education through data-driven insights. The system benefits both students and educators in the following ways:

- Students can share honest feedback about courses and teaching methods easily through an online platform.
- Lecturers and administrators can automatically analyze large amounts of feedback using AI, identify key issues, and receive smart curriculum improvement suggestions.

Although this project is developed for academic purposes, it has strong real-world potential and can be expanded into a complete institutional feedback management solution after deployment.

1.4 Project Overview

The AI-Powered Student Feedback Analyzer and Curriculum Suggestion System aims to build an easy-to-use web platform that helps universities and institutions collect, manage, and understand student feedback in a smarter way. Today, feedback is often collected and handled manually, which takes a lot of time and can lead to missing or misunderstood information. Our system will solve these problems by using automation and artificial intelligence (AI).

The main goal of this project is to make it easier for lecturers, coordinators, and administrators to gather feedback from students, analyze it quickly, and turn it into useful insights. The system will automatically find the overall opinion (positive, negative, or neutral), group similar comments, identify common topics (like teaching methods, assignments, or exams), and even suggest improvements to the curriculum based on what students say.

Included in the Project Scope:

- A web-based platform for creating, collecting, and managing student feedback.
- Secure login for administrators, lecturers, and coordinators.
- Tools to categorize, group, and summarize feedback.
- AI-based analysis of feedback sentiment and topics.
- Suggestions for curriculum improvement based on analyzed feedback.
- Visual dashboards and reports for better understanding.

Not Included in the Initial Scope:

- Mobile app version (only web-based for the first release).
- Integration with external systems or learning platforms (can be added later).
- Multilingual support and voice-based feedback input.

This project will improve how student feedback is collected and used in educational institutions. It will replace slow manual processes with an automated, smart system that helps teachers and administrators make better decisions and improve the learning experience.

1.5 Scope

This document explains what the AI-Powered Student Analyzer and Curriculum Suggestion System will do, its goals, and its limits. It will guide everyone involved in designing, building, testing, and launching the system.

The main aim of this project is to create a web-based tool that uses artificial intelligence (AI) to study student performance and feedback, understand learning patterns, and suggest better curriculum options. This will help teachers and educational institutions make smarter, data-based decisions to improve teaching and learning.

The project includes key parts such as tools for collecting student data, AI-based performance and sentiment analysis, automatic report generation, and easy-to-understand dashboards. It may also connect with existing Learning Management Systems (LMS) if needed.

However, this document does not include tasks like system maintenance after launch, creating third-party tools, or working with hardware devices. Any new features beyond what is listed here will be planned as future updates or separate projects.

1.6 Assumptions

- All student feedback will be given in digital text form (like typed comments or online survey answers).
- Students, teachers, and administrators will have basic computer skills and internet access to use the system.
- The educational institution will have the necessary setup (like servers, databases, and a stable network) to run the platform.
- Data privacy and security rules will be followed, and the institution will get all needed permissions before sharing feedback data.
- The system will be able to connect with existing Learning Management Systems (LMS) or other tools using standard methods like APIs.
- The AI language analysis will mainly support English feedback in the initial version.
- The feedback provided will be enough in amount and quality for the AI to learn and give accurate results.
- The system will only analyze and report feedback. It will not make or apply decisions for the institution.

1.7 Definitions, Acronyms and Terminology

Term / Acronym	Definition / Description
AI (Artificial Intelligence)	Technology that allows computers to think and learn like humans.
NLP (Natural Language Processing)	A part of AI that helps computers understand and analyze human language.
Sentiment Analysis	Finding out whether feedback is positive, negative, or neutral.
BERT (Bidirectional Encoder Representations from Transformers)	A pre-trained AI model is used to understand the meaning of words in sentences.
T5 (Text-to-Text Transfer Transformer)	An AI model that converts one form of text into another, useful for summarizing and analyzing feedback.
Machine Learning (ML)	A technique where computers learn from data and improve automatically.
Feedback Analyzer	The main part of the system that reads and understands student feedback.
Curriculum Enhancement	Improving courses and teaching methods using student feedback.
Dataset	A collection of feedback data used to train and test the AI model.
Dashboard	A screen that shows reports and analysis results.
Administrator	A person who manages users, data, and system settings.
Faculty / Lecturer	Teachers who use the system to review student feedback.
Student	The person who gives feedback about courses and teaching.
Report Generation	Creating summaries and reports from analyzed feedback.
Data Preprocessing	Cleaning and preparing feedback text before analysis.
Tokenization	Splitting text into smaller parts like words for easy processing.
Backend	The part of the system that handles data and AI operations.
Frontend	The part of the system users see and interact with.
Database	A place where all feedback and user data are stored safely.
UI (User Interface)	The design and layout users interact with.
API (Application Programming Interface)	Connects different parts of the system to share data.
Data Visualization	Showing analyzed data as charts or graphs for easy understanding.

2. PROJECT SCOPE AND IMPACT

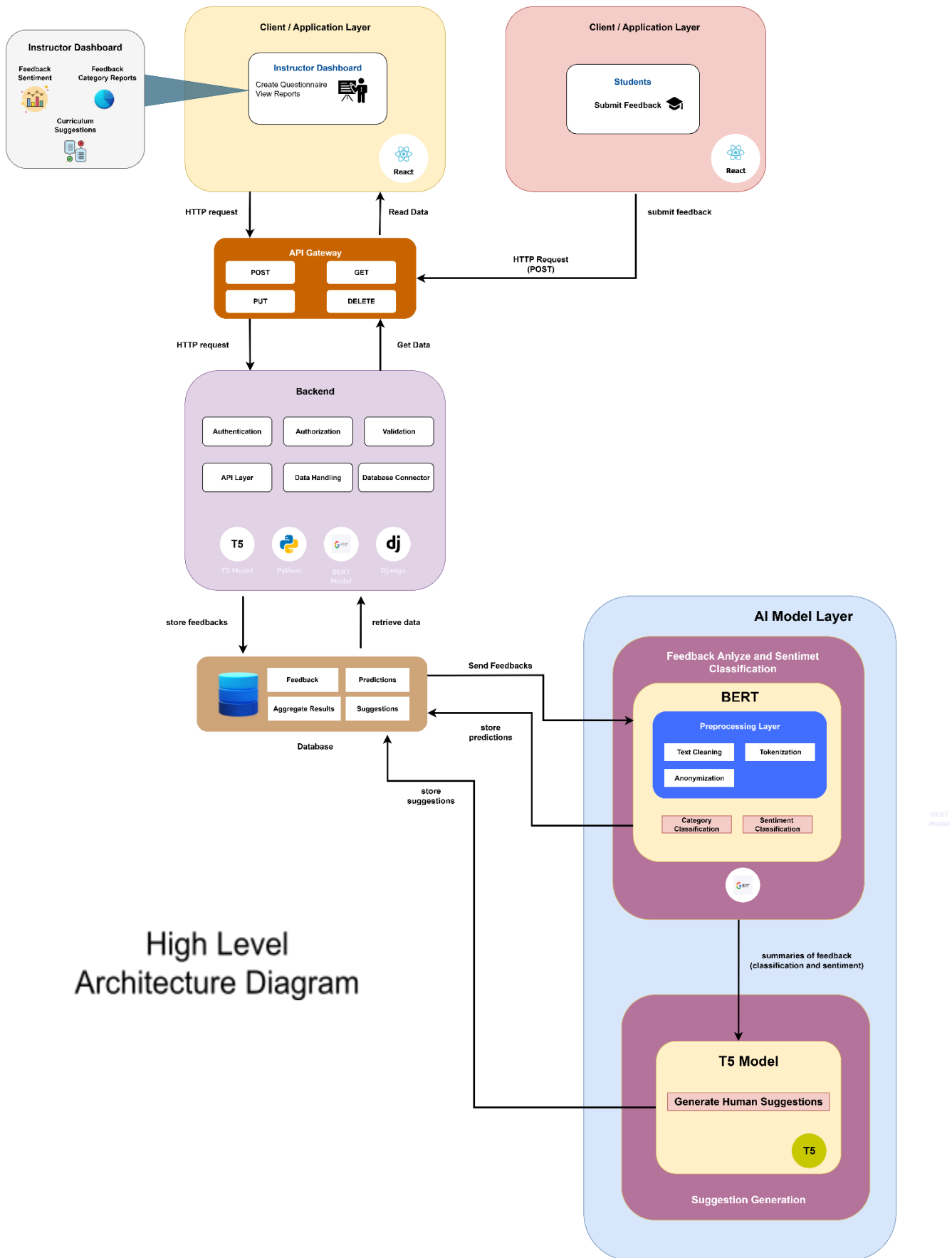
The main goal of the AI-Powered Student Feedback Analyzer and Curriculum Suggestion System is to make the feedback process faster, smarter, and more helpful for improving education. This system uses Artificial Intelligence (AI) to automatically read and understand student feedback, find common issues, and give useful suggestions to improve the curriculum.

With this system, educational institutions can:

- **Improve system performance:** Feedback will be collected and analyzed automatically, saving time and reducing mistakes that can happen with manual work.
- **Enhance service quality:** By better understanding what students

2.1 Scope Inclusions

- A web-based platform where lecturers can create and manage feedback forms.
- Automatic collection and storage of student feedback in digital format.
- AI-powered sentiment analysis to understand how students feel whether their feedback is positive, negative, or neutral.
- Topic detection and summarization to quickly identify common issues, suggestions, and important points from large amounts of feedback.
- Curriculum improvement suggestions are generated based on the analyzed feedback to help improve teaching and course content.
- Dashboards and visual reports that present feedback insights clearly for lecturers and administrators.



High Level
Architecture Diagram

2.2 Scope Exclusions

- **Mobile app** - The system will work only as a web platform in this phase.
- **Multi-language support** - Only English feedback will be analyzed in the initial version.
- **Chatbot support** - There will be no real-time chatbot to collect or reply to feedback.
- **Automatic curriculum updates** - The system will give suggestions but will not change the curriculum by itself.
- **Offline feedback collection** - All feedback will be collected and processed online only.

2.3 Impact on other systems

The Student Feedback Analyzer and Curriculum Suggestion System is designed to integrate seamlessly into the institution's existing digital ecosystem. It interacts primarily with academic management systems, user databases, and reporting tools. The project aims to minimize disruption to other systems while ensuring smooth data exchange for accurate feedback analysis and curriculum improvement.

2.3.1 Affected by Other Systems

- **Student Information System (SIS) / ERP System:**
Provides essential data such as student details, course enrollments, and faculty assignments. This ensures feedback is collected from verified students only.
- **Learning Management System (LMS):**
Provides data on assignments, course materials, and teaching schedules. This data helps contextualize feedback, allowing the AI model to correlate comments with academic performance indicators.
- **Authentication System:**
Integrates with institutional login systems (e.g., LDAP, OAuth, or Single Sign-On) to verify user credentials securely.
- **Institutional Database:**
Supplies static data such as course codes, department details, and semester information necessary for mapping feedback correctly.

2.3.2 Affects on Other Systems

- **Academic Administration System:**

Receives summarized AI-generated reports and suggestions for curriculum updates. Administrators may require new dashboards or modules to view this data.

- **Course Management System (CMS):**

May be updated to display summarized student feedback statistics or integrate suggestion insights for faculty members.

- **Reporting and Analytics Systems:**

The AI-generated suggestions and sentiment summaries may be exported to institutional analytics tools for strategic decision-making. These tools might need minor configuration changes to accept new data formats (e.g., CSV, JSON).

- **Database Maintenance Tools:**

Backup and archival systems may require updates to handle the new “feedback” and “suggestion” data tables, ensuring proper retention and compliance.

3. FUNCTIONAL REQUIREMENTS

This AI-Powered Student Analyzer and Curriculum Suggestion System functional requirements define what a system should do and how it should operate in different conditions. The system's main functions include feedback collection, sentiment and topic classification, curriculum suggestion generation, reporting, user management, data archival and administration operations. Additionally, this system must handle unusual conditions carefully by providing error handling, input validation, recovery procedures and secure communication.

3.1.1 Function

3.1.1.1 Function 1 - Student Feedback Collection

The most important function of this system is to collect feedback from students. This module mainly allows the students to submit their feedback about course infrastructure. Such as lectures, assignments, day school, labs, exams, course materials, and an online learning platforms.

This feedback collection process is structured format. Structured data provides clear insights. Free text comments provide context for AI driven analysis.

Detailed Description of Student Feedback Collection

- First, Administrator of the University who is represent the particular University. He is created the user account for university in this system. After enter the system, he gives the permission to coordinator or lecturer to create the feedback form.
- After that, lecturer or coordinator creates the course section to attach the questionnaire (customizable questionnaire). As well as a lecturer or coordinator will add the student information who is register that course after create course section.
- After that, Lecturer or coordinator created customized questionnaire (it customizable web form developed using React) through the system. After created questionnaire, the link(tokenized link) will be generated. So they copy the link and will put into student communication medium like LMS.
- When student will click the link, it will navigate to that questionnaire form site. It will not directly allow to the student fill the questionnaire. Instead of, it will ask the student

ID. Student after give the student id, system verify that student id and will show the register courses of the student.

- Then student can click any register courses and can provide the feedbacks. Submitted feedback is anonymized to ensure student privacy and stored in the database along with meta data. Such as timestamp, course and category.

3.1.1.2 Function 2 - Feedback Analysis

Here, planned to use the BERT Model. Here System preprocessed the collected text data. Such as tokenization and cleaning. Here BERT model needs to be fine-tuned to this system accordingly.

Detailed Description of Feedback Analysis Process

- After student submit the feedback form it will store the database.
- but sometimes some students do not fill the feedback so we can not set as no of students who registered the course as trigger point that means we can not set after all students give the feedback then text will be analyzed. so we planned to set minimum students instead of all registered students of the course as a triggering point that means we planned to analyze the feedback form after minimum students of the course responded.
- After minimum students responded the form, feedback text will be preprocessed. In this preprocessed text process, the data will be converted to lowercase, remove the punctuation in the sentence, remove stop words like “the”, “and”, then change the words like simple grammar format. For this purpose, we can use NLP techniques.
- Here planned to use BERT pre trained model for text analyze and will train that model.
- As well as are not going to directly use that pre trained model instead we planned to customized the model as for the system requirements because BERT is pre trained on general English language from different sources like books and wikipedia but student feedback is very specific domain. It includes informal writing, mixed sentiments and education related terms. So, we fine-tune BERT on our labeled dataset of student feedback so that it learns the structure, tone, and vocabulary specific to our use case. This improves the model’s ability to predict sentiment and categorize feedback accurately.

- It understand the emotion of the words which get in the feedbacks using pretrained models (BERT). For example, whether the sentence is positive or negative or neutral.
- After that, system will find the common topics. That is, what topics student are mostly talk about. Example teaching or assignments or exams. summary of the feedback also generated.
- Then group similar comments that gave from students.

3.1.1.3 Function 3 - Suggestion Generation

As well as provide suggestions to the lecturer or coordinator or administrator to improve the curriculum. For this, planned to use the T5 model. T5(Text to Text transfer transformer) generates human readable suggestions.

Detailed Description of Suggestion Generation

- First should prepare datasets. Then fine-tuned T5 system with prepared datasets.
- When giving the input from the aggregate result from BERT, it generates suggestions and sends them to the lecture interface. This means the backend aggregates BERT classifications outputs into human-readable summaries, constructs a standardized prompt and request suggestion from T5 service.
- The T5 model returns a natural language curriculum recommendation. The lecturer can see feedback and suggestions for a particular course in the dashboard.

Example:

Input: “70% students negative on assignment”.

Output: “Reduce assignment workload and provide more example to support students”

3.1.1.3 Error handling

- If the students tries to skip the mandatory questions in the form, an error message will show about missing response.

3.1.2 System Administration

The System Administration module is responsible for managing for overall functioning, configuration, and maintenance of the student feedback analyzer. Administrators will have the highest level of advantages and will be able to see all data, user accounts and application settings. Here, lecturers or the university staff are also a type of administrator. So they can add, update or remove the feedback questionnaire and manage course details such as course name, course code, student participation and instructor permission.

Here, first Administrator of the University who represents the particular University will be created the user account for University in this system. After enter the system, he gives the permission to coordinator or lecturer to create the feedback form. After that, lecturer or coordinator create the course section to attach the questionnaire (customizable questionnaire). As well as a lecturer or coordinator will add the student information who is register that course after create course section. So here Lecturer or administrator of the university is one type of administrator of this system.

Another important of this system is the main administrator who is responsible for managing and maintaining this system. They are responsible for performing database maintenance task such as cleaning outdated records, archiving past semester data and optimizing indexes for performance. As well as they will manage machine learning model updates such as uploading new finetuned versions BERT and T5, configure relation policies and handle error logs for troubleshooting.

In this system, the university and its staff are involved so security is very important factor so system requires strong authentication, role base access and activity auditing. As well as here different roles are involved such as administrator, lecturer and university staff so role base access implementation is very important.

As for database administration, university creates many courses for getting feedback so manage course table, questionnaire templates and feedback responses are very important.

3.1.3 Data Archival and Retention

The Data Archival and Retention function ensures that all feedback related information, model generated insights and system activity logs are securely preserved, organized and retrievable for future academic cycles. In this system, feedback data is collected at the end of every semester from multiple courses, and analyzed through BERT and T5 based models and then archived in a structured manner.

When the semester ends, all finalized feedback records, aggregated summaries, model outputs such as sentiments, categories and suggestions and instructor evaluations are marked as semester ends. The system automatically transfers these datasets from the primary operational database to the designated archival storage environment.

Archived data plays a critical role in continuous improvement. During the next academic cycle, this archived data will be used for comparative analytics to evaluate changes in student satisfaction, identify persistent issues and refine course materials. For example, archived feedback and model-generated suggestions from course “EEY6189-2025” may guide notifications to course “EEY6189-2026”, ensuring the course evaluation process.

Students' anonymized feedback data, analysis results and generated reports are kept for a defined duration. After a specific period, older records are securely removed following institutional and data protection regulations. The system enforces access control on archived data, ensuring authorized administrators or people can retrieve and review past records.

In addition, the system periodically performs integrity checks and maintains audit logs verify archived datasets remain unaltered and retrievable. The well structured archival process supports institutional reviews, data driven decision making and model retraining with historical data.

3.1.3.1 Archival Workflow

1. Triggering Condition

At the end of each semester, when feedback collection is marked as complete, an archival job is triggered automatically by the backend schedule.

2. Data Extraction

Relevant tables are exported into structured archival schemas like JSON, SQL, grouped by academic year, semester and course code.

3. Data Validation

Before archival, the system performs integrity checks such as row counts, foreign key and consistency and ensures that no data corruption exists.

4. Archival Storage

Archived data moved to database. Compressed archives are saved in secure cloud storage.

5. Indexing

Metadata indexes are created for quick lookup(by course, semester, faculty id) to enable selective retrieval of past datasets.

6. Access Management

Only system administrators, academic coordinators and authorized faculty can access archive datasets. Access is controlled through role-based permissions.

3.1.3.2 Data Restoration and Retrieval

- The system provides an admin interface or API to retrieve historical archives for comparative analysis or reporting.
- Retrievals are logged with timestamp, user id and purpose to ensure transparency and accountability.
- During retrieval, archived dataset can be reloaded into a temporary analysis environment for visualization or retraining machine learning models.

3.1.3.3 Data Protection and Security

- All archived files are encrypted.
- Audit trails are maintained for every read, write or delete operation performed on archived data.

3.1.3.3 Error Handling and Recovery

- If archiving fails in the system,
 - Logs the failure with reason and stack trace.
 - Notify the system administrator via laert or email.
- Restoration failures are similarly logged, with a rollback procedure to avoid partial recovery.

3.1.4 User Profiles, Roles and Privileges

Different users interact with the system and each user must have specific privileges to ensure security and smooth operation. Here role based access control will be implemented. As well as if any unauthorized access attempts occur, system trigger alerts nd logging.

1. Students

- Can submit feedback about courses they are learning.
- Can not view other students' feedback.
- Can see their registration course feedback forms.

2. Lecturer

- Can view summarized reports for their own courses.
- Can access T5 model-generated curriculum improvement suggestions.
- Can see the number of responses they got.

3. University Administrator

- Administrator who represents the particular University can create the account in the system and give access permission to lecturer for creating the questionnaire and access the feedback.

4. System Administrator

- Have full access including user management, database maintenance, AI retraining and system monitoring.
- As well as he will responsible person for managing archival, data integrity and reporting.

Functionality Table

Functionality	Student	Lecturer	Admin
Submit feedback	✓	✗	✗
Edit Feedback	✓	✗	✗
View Confirmation	✓	✗	✗
View summary reports	✗	✓	✓
Export Report	✗	✓	✓
Manage Users	✗	✗	✓
Manage Courses	✗	✓	✓
Manage Questionnaire	✗	✓	✗
Archival	✗	✗	✓

3.1.5 Reporting Requirements

One of the core outputs of the system is generation of reports and summaries based on feedback analysis. Reports must be interactive, exportable and easy to understand for lecturers and administrators.

Detailed description about reports

- Sentiment Analysis and Category wise Report: It shows the percentage of positive, negative and neutral responses. As well as highlight issues in teaching, labs, assignments, exams, course materials and platform.
- Curriculum Suggestions: It shows the suggestion using T5 models.

4. NON-FUNCTIONAL REQUIREMENTS

4.1.1 Performance and Load Requirements

Current User Load	The System is expected to handle 100-200 concurrent users (students and faculty) during feedback periods.
Expected Growth	
Number of concurrent Users	This system must support up to 500 feedback submissions concurrently in peak usage scenarios.
Transaction Size	Feedback Text Responses typically range from 50-200 words and stored as less than 10KB.
Maximum Average Transaction Time Acceptable	The system should show acknowledgement of feedback submission within 2-4 seconds.
AI model Processing Time	Sentiment and suggestion generation should complete within 8-10 seconds per batch of feedback.
Server Throughput	Backend API should handle at least 50 requests per second under load testing.
Database Query Time	Retrieval and analysis results should not exceed 2 seconds for standalone queries.

4.1.2 Compatibility Requirements

HTML Versions to be supported	HTML5, CSS3 and Javascript ES6, REACT JS
Backend Environment	Python 3.10 or higher and Django Framework
Browser Versions to be supported	Google Chrome, Mozilla Firefox, Microsoft Edge
Database version to be supported	PostgreSQL 13+
Communication Protocol	RESTful API over HTTPS
Platform versions to be supported	Windows 10 or higher
External System or Libraries	Hugging Face Transformers for AI models(BERT, T5), JSON-based data exchange between frontend and backend

4.1.3 Certification Requirements

Here, no formal third-party certifications are required. However, the system shall follow with following standards.

- Data Privacy and Security Compliance: System follow to institutional data protection guidelines similar to GDPR and FERPA principles.
- Code Quality Standards: Python PEP8 and W3C compliance for frontend code.

4.1.4 External Interface Requirements

Our AI-based feedback analysis system will support communication between the web application, AI service APIs, and the database. External interfaces include RESTful APIs for AI model interaction, and secure HTTPS connections for data exchange between client and server.

4.1.4.1 User Interface

The application will provide an interactive, web-based user interface accessible through modern browsers. It will include dashboards for lectures and administrators and feedback forms for students. This interface will follow responsive design principles for desktop, using simple navigation, clear visualisation, and role-based access.



Figure 1 : Landing Page



Register Your Institution

Create your account to start collecting and analyzing student feedback using AI-powered insights for curriculum enhancement

Create Institution Account

Fill in the details to register your institution

Institution Information

Institution Name

Name of the Institution

Institution Type

Select type from dropdown

Department/Faculty

Department name

Official Website

https://www.abc123.lk

Address

Country / Region

Select Country

Coordinator Details

Full Name

Your full name

Designation

Select designation

Official Email

official@email.com

Phone Number

077-----

Security

Password

Create password

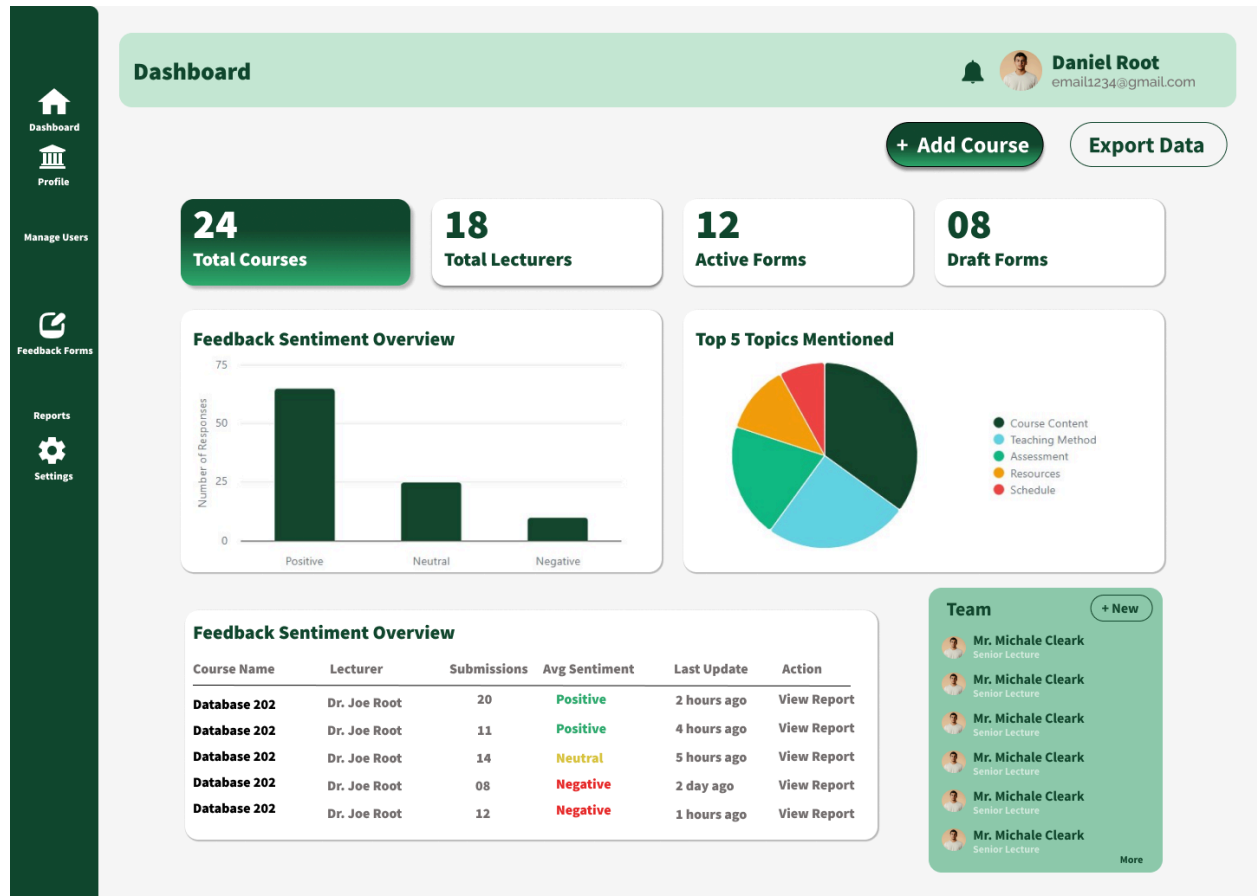
Confirm Password

Confirm password

☐ I confirm that I am an authorized representative xxxxxxxxxxxx

Register Institution

Figure 2: Institution Registration Page



Course Information

Course Name

Enter the official course title

Course code

Enter the unique course code

Department/Faculty

Department name

Semester/Academic

The time period when the course is offered

Description

Any Note

Coordinator name

Name of the course coordinator

Lecture Name

One or more lecturers teaching the course

Reset

Create Course



Create new course

Create your account to start collecting and analyzing student feedback using AI-powered insights for curriculum enhancement

Figure 4 : Course create page

Feedback Form

Build a custom feedback form to collect insights from students in this Course

Save Draft

Publish Form

Form Information

Course Name

Information Security 003

Form Title *

e.g Mid-Semester Feedback

Description

Provide context about this feedback form..

Feedback Type

Information Security 003

Add Questions

0 questions added

1

Question Text

Enter your question.....

Question Type

Short Answer

☐ Required Question

+ Add New Questions

Form Preview

Preview Form

Preview the form

Quick Stats

Total Questions	1
Required Questions	0
Estimated Time	2- 3 Min

Save Draft

Publish Form

Figure 5 : Feedback form page

4.1.4.2 Hardware Interface

- The System runs on standard web-enabled devices such as PC, laptops, tablets and Mobile Phones.
- A Standard Server is required for hosting.
- The system needs a GPU for training the models, so planned to use Google Colab during the training phases.

4.1.4.3 Software Interfaces

- For Front-end and Backend communication, using RESTful APIs over HTTPS with JSON format.
- For Backend and Database communication, using Object relational Mapper. Example: Django ORM.
- For Backend and AI model communication, integrate python via Hugging Face Transformer APIs for BERT and T5 Models.
- Planned to integrate with PyTorch, NumPy, pandas and scikit-learn for Machine Learning operations.

4.1.4.4 Communication Interfaces

- Communication between all system components occurs via secure HTTPS (SSL or TLS) protocols.
- The System supports API versioning for future upgrades.
- Backend endpoints are accessible through port 443 (HTTPS).

4.1.5 Security and Authentication requirements

4.1.5.1 Data Storage Security

- All user credentials and feedback data are stored in an encrypted form.
- Database encryption for sensitive information.
- Regular backup and recovery mechanisms.
- Access restricted to authorized roles only.

4.1.5.2 Data Communication Security

- All communications between the frontend, backend and AI components occur via SSL-secured HTTPS.
- Use of POST methods for data submission to prevent data exposure in the URL.

4.1.5.3 Authentication

- Using Role-based authentication (Student, Lecturer, Admin).
- Secure login using username and password.
- Session timeout and token expiration for security.

4.1.6 Quality Assurance Requirement

4.1.6.1 QA Test Scope

- Unit Testing: For backend APIs and AI model components.
- Integration Testing: ensures smooth communication between frontend, backend and API layers.
- System Testing: Full end-to-end verifications of workflows.
- Performance Testing: checks response time and scalability under load.
- User Acceptance Testing: Conducted by faculty or students to validate usability.

4.1.6.2 QA Environment

- As for software, can use Postman(API Testing) and Python(PyTest).
- As for hardware, standard test machines and optimal GPU are needed for AI performance tests.
- As for Data, uses anonymized test feedback data for safe evaluation.

4.1.6.3 QA Data

- Include pre-labeled student feedback datasets for training or testing the AI models.
- Synthetic data generation tools may be used to expand dataset diversity.
- Validation data is stored in a separate sandbox environment to prevent contamination of production data.

4.1.7 Development Requirements

4.1.7.1 Development Environment

Software Requirements:

- Front end: React.js with Tailwind CSS
- Back end: Python framework Django
- Database Layer: Postgre SQL
- AI & NLP models: BERT for sentiment analysis and T5 for suggestion generation
- Language Processing Frameworks: Hugging Face Transformers.
- Data Visualization Tools: Matplotlib
- Hosting: Namecheap Server
- Version Control: Git and GitHub

Hardware Requirements:

- Require standard local computers for coding, integration and testing.
- Model fine-tuning is performed using Google Colab it provides GPU virtual machines for training and testing AI models.

Development Tools:

- Visual Studio Code

4.1.7.2 Development Data

Development process requires well-structured datasets to train and fine-tune the AI models.

Key Data Entities:

- Student feedback Text
- Sentiment Labels
- Topic Categories
- Suggestion

4.1.7.3 Coding Standards

Code should be follow language conventions, naming conventions, documentation and versioning standards to maintain code readability, consistency and quality

4.1.7.4 Implementation Packaging Requirements

Application will be modular, separating the front end, back end and AI modules tools like Vite and npm will be used for packaging and dependency management.

4.1.8 Deployment Requirements

4.1.8.1 Installation Packaging Requirements

- System will be packaged for installation in both development and production environments using containerization tools.
- The packaging tool for application containerization is Docker.

4.1.8.2 Deployment Requirements

- Our application will be deployed on Namecheap server hosting.
- HTTPS protocol with SSL certificate

4.1.8.3 Documentation Requirements

Needed deployment documentation will be provided to ensure smooth setup and usage.

- Installation Guide
- Deployment Manual
- AI Services configuration guide
- System user manual

4.1.9 Internationalization requirements

System is initially designed for English-speaking users but is prepared for future multilingual support.

4.1.10 System Activity log requirements

Application must maintain detailed activity logs for auditing, troubleshooting, and security tracking.

Types of logs:

- User Activity Logs: Login/logout, form submissions, data exports, and role-based access actions.
- System Events: API calls, database updates
- Error Logs: Server errors, failed processes, AI model failures and unauthorized access attempts.
- AI model logs: Record of model inference requests, processing time and prediction summaries.

4.1.11 Special Documentation Requirements

- The system and its related documents must include necessary legal disclaimers and ownership information to protect property and maintain compliance.
- Trademarks and logos are used under consider the branding policies.

4.1.12 Applicable Standards

- ISO 9001: Ensure performance
- IEEE 830-1998: Defines SRS structure and best practices
- GDPR: Ensures confidentiality of student data
- WCAG 2.1: Supports inclusive access for all users
- HTTPS, TCP/IP: Secure client-server communication

4.1.13 Licensing Requirements

- MIT License: Allow free use, modification, and redistribution for educational and research purposes.
- Usage Restriction: The software may not be used for commercial purposes without permission from the development team.

4.1.14 Online User Documentation and Help System Requirements

- A built in Help section will explain how to submit feedback, view reports, and use analysis tools.
- Step by step guides and FAQs will be available inside the system.
- Tooltips will appear when users hover over buttons or icons to explain their functions.
- A simple Help & Support page will include contact details or a form for technical help.
- All help content will be written in clear, easy-to-understand language.

4.1.15 Purchased Component

- **AI/NLP Tools:** APIs like Google Cloud NLP will be used for feedback analysis.
- **Database System:** MySQL or PostgreSQL will store feedback data securely.
- **Cloud Hosting:** AWS or Google Cloud will host the system for reliability and scalability.
- **UI Frameworks:** Tools like React or Tailwind CSS will be used to create a clean web interface.

4.1.16 Technical Requirements.

- The system will use Python for the backend and AI, and HTML, CSS, and React.js for the frontend.
- MySQL will be used to store and manage data.
- Using the BERT NLP Model for feedback analysis and to understand the sentiments. As well as using the T5 model for suggestion generation.
- The system will work smoothly on browsers like Chrome, Edge, and Firefox.
- It will run on both Windows and Linux servers and support up to 100 users at once.
- Cloud hosting ensures speed and scalability, while data encryption and backups protect user data.

4.1.17 Supportability Requirements

- The code will follow standard naming and coding practices for easy maintenance.
- The project will use Git/GitHub for version control and collaboration.
- Logs and documentation will help developers fix issues quickly.
- Regular maintenance and updates will keep the system stable and secure.

4.1.18 Reliability Requirements

- The system should work reliably with at least 99% uptime.
- Auto-save and backup features will prevent data loss.
- Built-in error handling will detect and fix small issues automatically.
- Regular performance testing will ensure smooth operation for many users.

4.1.19 Usability Requirements

- The interface will be simple, clean, and easy to use for students and staff.
- It will work on all major browsers and adapt to different screen sizes.
- Buttons, menus, and text will be clear and well-labeled.
- Usability testing will be done to make sure the design is easy to navigate.

5. PILOT SUCCESS CRITERIA

The success of the pilot implementation will be evaluated based on the improvements in efficiency, accuracy and user satisfaction compared with the existing manual feedback process.

Success Criteria	Measurement Method	Expected Outcome
Time reduction in feedback Processing Time	Compare the time required to manually analyse student feedback vs AI-based system.	At least 70% reduction in total analysis time
Improvement in sentiment classification accuracy	Evaluate the accuracy by comparing the AI results with manual evaluation by lecturers.	Equal to or greater than 85% accuracy in detecting positive, neutral, and negative sentiments.
Improved topic detection	Compare the number of meaningful themes identified manually vs AI system output.	AI system identifies at least 90% of relevant themes reported by human evaluators.
Lecture and administrator satisfaction	Conduct surveys before and after the pilot use to measure satisfaction with feedback analysis.	At least 80% user satisfaction with system usability.
Student participation rate	Compare the feedback submission rates before and after system deployment.	20-30% increase in the number of students submitting feedback forms.
Reduction in Manual workload	Track average hours spent by lecturers.	At least 50% decrease in manual effort for report generation.

6. APPENDIX

i) Flow Diagram of the System

